EpistemoBrain: Generalizing Executive Memory to Multi-Agent Systems

Zhuqi Ma[1] **Haopeng Cui**[2,†]
[1]Renmin University of China
[2]Tsinghua University
2025201699@ruc.edu.cn chp25@mails.tsinghua.edu.cn
† Corresponding author

Abstract

Multi-agent systems built on large language models now carry long-horizon tasks across many role-specialized agents, and they run into the same bounded context that a single agent runs into. Executive memory was introduced to hold one reasoning agent's trajectory together under that budget: a dependency-aware graph of its reasoning steps, with Flush, Fold, and Recall to prune invalid paths, compress completed sub-trajectories, and rebuild a compact, high-salience context. Every part of that mechanism assumes that the agent writing the memory is also the agent reading it. We ask what becomes of the mechanism once that assumption fails. **EpistemoBrain** generalizes executive memory from one agent to an agent collective. The unit of analysis shifts from a private trajectory to a shared cognitive state, and each of the three native operations changes meaning with it. Flush becomes *Quarantine*, so a refuted claim keeps its record instead of vanishing. Fold becomes *Dialectical Fold*, so a dispute compresses only after both strongest positions, the hinge variables, and the falsification conditions have been preserved. Recall becomes *Perspective Recall*, so each agent reconstructs a different projection of the shared state instead of reading one undifferentiated summary. Two constraints follow that the single-agent setting never raises. A lone agent's memory may be private and freely prunable; a collective's memory is public and has to stay auditable. And process has to separate from committed knowledge, because in a collective a working guess is readable by peers who will take it for verified unless the two are physically distinct. Those constraints carry the rest of the design: two graphs joined by an append-only ledger with one database-enforced writer, an admission gate that every formal change clears before it becomes a conclusion, and an aggregation rule that forbids majority voting and returns conditional consensus instead. None of it is tied to a task domain. The mechanism is defined over a set of role-specialized agents, a shared committed graph, and an admission policy, and those three are the only things an implementer supplies.

We instantiate EpistemoBrain as a scientific controversy review, where a confident wrong answer is expensive: a fluent, well-cited summary can hide a causal claim built on cross-sectional correlation, six papers that draw on a single dataset, or a dissenting view that the summarizer quietly averaged away. The instantiation organizes seven specialist agents through a formal protocol (independent precommitment, professional acquisition, evidence exchange, cross-examination, blindspot bounty, joint modeling, and final rejudgment) and returns an *auditable controversy evidence map*: conclusions sit next to their limitations, every claim traces to its source, and no dissent is deleted. Genuine disagreement survives as a named Dissent Certificate. We evaluate it on the **Foresight-Blindspot** benchmark, a time-sliced evaluation protocol for blindspot discovery in computational social science. The study combines a controlled comparison of five baselines (single-agent deep research, fixed multi-agent debate, a collective with linear context, a collective with executive memory but no evidence engine, and the full system), a six-variant ablation ladder that removes one capability at a time, and repeated runs against a real model endpoint. Coverage tracks agent coverage: a lone agent recovers one of seven gold blindspots, while every collective configuration recovers all seven, at 1.8 times the ledger volume and roughly a quarter of the cost per blindspot. The ablations locate which roles carry coverage, since removing the adversarial falsifier or the evidence auditor costs a gold blindspot, and they expose a self-report trap: strip lineage tracking and the system's own evidence-independence score rises to a perfect 1.0. On the real endpoint the full system is the only configuration that reaches perfect blindspot precision with every honesty signal intact, including dissent preservation at 1.0 across all fifteen runs. The price is recall, because the same protocol leaves 12.3

structured slots unfilled on a flash-tier model where every other variant leaves between one and two. We report that trade-off as a deployment boundary instead of smoothing it into a single score, and we ship the harness that measures it.

1 Introduction

LLM agents increasingly work in teams. Multi-agent debate sharpens reasoning (Du et al., 2023; Liang et al., 2023), role-specialized panels cover ground that a single prompt does not reach (Chen et al., 2023), and long-horizon pipelines hand retrieval, analysis, and synthesis to separate agents (Qian et al., 2024). Teaming solves a capability problem and creates a memory problem. A collective that runs for hours accumulates far more reasoning than any member can hold, and its members cannot simply share one transcript: the transcript grows without bound, and handing every member the same text gives them an undifferentiated view of work that only some of them did.

The obvious remedies carry three failure modes that a fluency-optimized summary cannot repair. Compression that discards whatever looks settled deletes the wrong material, because a lone reader tends to see only its own interpretation: the measurement specialist's concern about self-report bias is invisible to a pipeline that has no measurement specialist, and it is the concern that gets folded away. Agreement is not evidence either. If several agents read the same erroneous abstract, the same retracted paper, or the same misattributed citation, the system can package a *shared error* as a consensus that looks verified simply because it was repeated (Ioannidis, 2005; Open Science Collaboration, 2015). Finally, a summarizer's closing "everyone basically agrees" step erases exactly the information a collective's memory has to preserve: what the dissenting agent opposed, and which evidence would change its mind. These are not prompt-tuning problems. They are structural questions about how a collective's memory should be organized.

Computational social science makes the stakes concrete, and it is the domain we adopt as our reference instantiation. The reproducibility crisis has shown that a large share of published findings do not replicate (Open Science Collaboration, 2015), that publication bias inflates apparent effect sizes (Fanelli, 2010), and that researcher degrees of freedom multiply false positives (Gelman and Loken, 2013; Nosek et al., 2018). In the domain this work targets, digital behavior, social media, and mental health, cross-sectional designs are routinely reported as causal, self-report measures dominate, and a small number of datasets support a large fraction of the literature (Orben and Przybylski, 2019; Valkenburg et al., 2022). An evidence reviewer for this domain needs more than an accurate summary. It has to *audit* claims against their sources, *count independent evidence* rather than papers, and *keep the dissent* visible.

Long-horizon agentic work already has a memory answer for one agent. **MemoBrain** (Qian et al., 2026) gives a single reasoning agent an executive memory: a dependency-aware graph of its reasoning steps, with Flush, Fold, and Recall to prune invalid paths, compress completed sub-trajectories, and rebuild a compact, high-salience context under a fixed token budget. Every part of that mechanism assumes one agent. One agent retrieves, one agent reasons, and one agent compresses its own past before the budget runs out. Our starting question is what the same mechanism has to become when the one agent turns into a collective.

We propose **EpistemoBrain**, a domain-general executive memory and governance layer for multi-agent systems. It generalizes executive memory from a single agent to an *agent collective*: a set of role-specialized agents that share one committed knowledge graph, one admission policy, and one protocol. Nothing in the mechanism names a task domain. An implementer supplies three things (the role specifications, the admission policy, and the schema of the committed graph), and the same operations apply; the reference instantiation we describe and measure later is one such supply, not the definition. EpistemoBrain sits above the agents. It is not an extra member and it holds no vote. It manages their long-horizon process, preserves the counterexamples and unresolved conflicts they produce, turns the state of what they have committed into the next round of work, and holds the line between process memory and committed knowledge.

The design starts from the opposite direction of "several chatbots plus a summarizer": we first decide what memory and organizational structure a reliable collective requires, and only then place the roles inside it. Four commitments follow. Every agent participates in every task, with cost held down by speaking budgets, semantic deduplication, and shared retrieval caches instead of by dropping agents. Dissent is a first-class object: majority voting is forbidden, outcomes are conditional consensus, and an agent that cannot accept the result receives a named certificate. Process and committed knowledge live in separate graphs, coupled by an append-only event ledger and a single database-enforced writer. Memory is layered and public: private per-agent process memory, collective executive memory, and the committed graph. The three native operations are then redefined as Quarantine, Dialectical Fold, and Perspective Recall.

We evaluate the reference instantiation on **ForesightBlindspot**, a time-sliced evaluation protocol for blindspot discovery in computational social science. The evidence comes from a controlled comparison of five baselines and a six-variant ablation ladder on a scripted case with no model calls, no network, and no database, together with end-to-end verification on real research tasks. Blindspot coverage follows agent coverage: after we corrected the single-agent baseline's role-selection semantics (Section 4.2), it finds one of seven gold blindspots while every collective variant finds all seven. The ablations locate the roles that carry coverage, since removing the adversarial falsifier or the evidence auditor costs a gold blindspot, and they expose a self-report trap: strip lineage and the system's own evidence-independence score inflates. Under scripted gateways the remaining mechanisms (memory, admission gate) leave no trace on the blindspot metric, which we record as a boundary of the measurement and not as a claim about what those mechanisms do.

Our contributions are threefold: (1) We *generalize executive memory from a single agent to an agent collective*, domain-independently. The unit of analysis moves from a private reasoning trajectory to a shared cognitive state, and we give the three native operations collective semantics: Quarantine, Dialectical Fold, and Perspective Recall. Section 3.2 states what breaks when a single agent's memory graph is reused unchanged, and why the generalized mechanism is forced to separate process from committed knowledge. (2) We design a *memory-governed collective protocol* around that mechanism. Agents interact through structured actions (propose, support, challenge, qualify, fork, request, revise, dissent) under an explicit prohibition on majority voting, producing conditional consensus with minority positions preserved. Every formal change is governed by an append-only, idempotent, replayable event ledger, a multi-stage admission gate, and a single projector whose write privileges are enforced at the database privilege level. (3) We instantiate the mechanism as a scientific controversy review and release the implementation, the **ForesightBlindspot** evaluation protocol, a five-baseline comparison, and end-to-end verification, establishing a measurement basis for blindspot discovery in computational social science.

2 Related Work

2.1 Multi-Agent Debate and Collaboration

Multi-agent debate improves reasoning by having agents critique one another's answers (Du et al., 2023; Liang et al., 2023; Chen et al., 2023; Qian et al., 2024), and role-specialized panels have been proposed for domain reasoning. These systems typically close with a summarization step that aggregates positions, and none addresses the failure mode we consider central: the summarizer can flatten a genuine dissent, and the agents can converge on a shared evidence error that looks like independent verification. EpistemoBrain differs on both axes. It forbids majority voting as a truth mechanism, and it subjects the material an agent wants to commit, not the arguments themselves, to an admission gate before anything enters the shared graph.

2.2 Memory for Long-Horizon Agents

Long-horizon reasoning strains bounded contexts, which motivates explicit memory architectures (Packer et al., 2023; Sumers et al., 2023). The closest predecessor to this work is **MemoBrain** (Qian et al., 2026), which gives a single tool-augmented agent an executive memory: a dependency-aware graph of its reasoning steps, with Flush, Fold, and Recall to prune invalid steps, compress completed sub-trajectories, and preserve a compact, high-salience reasoning backbone under a fixed context budget. EpistemoBrain generalizes that mechanism to an agent collective. The unit of analysis moves from one agent's trajectory to a community's shared cognitive state, which forces each native operation to change meaning: Flush becomes Quarantine, so a refuted claim keeps its status history; Fold becomes Dialectical Fold, so a dispute compresses only once both strongest positions, the hinge variables, and the falsification conditions are preserved; and Recall becomes Perspective Recall, so each agent recalls a role-specific projection of the shared graph. The generalization also raises a requirement the single-agent setting never faces. A collective's memory is read by agents that did not produce it, so process memory and committed knowledge have to be separated into two graphs; one agent's momentary guess must never quietly upgrade into a conclusion its peers treat as verified.

2.3 Retrieval-Augmented Generation and Factual Verification

RAG grounds generation in retrieved documents (Lewis et al., 2020), and verification pipelines check claims against evidence (Thorne et al., 2018; Wadden et al., 2020; Min et al., 2023; Dhuliawala et al., 2023). These systems verify *textual entailment*. Auditing a *scientific claim* requires checks that standard entailment systems do not perform: evidence levels (full text with exact quotes versus abstract-only metadata), citation entailment (an introductory hypothesis must not be reported as a tested result), causal-upgrade policy (cross-sectional correlation must not become a causal claim), evidence independence (papers sharing a dataset count as one independent evidence cluster), and source provenance (retraction status, preprint lineage). We reuse the production rules for these audits in our scoring functions instead of re-deriving parallel approximations.

2.4 LLM Agents for Science

LLM agents are increasingly applied to scientific tasks, from hypothesis generation to literature analysis. A recurring concern is that agentic systems can produce authoritative-sounding claims without grounding (Ioannidis, 2005; Open Science Collaboration, 2015). Our response is to make grounding a governance property rather than a prompt property, and to do it in a way that does not depend on the domain: the three operations, the dual-graph separation, and the non-voting aggregation rule are stated over agents, a shared graph, and an admission policy, and science supplies only one set of values for those three. We are not aware of a memory architecture that generalizes a single agent's executive memory to a collective, separates process from committed knowledge at the database privilege level, and ships a time-sliced protocol for measuring what the collective failed to notice. To our knowledge, EpistemoBrain is the first.

3 Method

3.1 Overview and Formalization

Figure 1 gives the architecture. A task is submitted as a contract: the objective, scope, budget, and optional materials. The mechanism runs the collective through its deliberative rounds, appending every formal change as an event to the event ledger. The Graph Projector, the ledger's only consumer and the committed graph's only writer, applies the admission gate to each candidate and materializes the admitted nodes and edges. Gaps the graph exposes generate bounties, which are scored, ranked, and broadcast back into the collective (epistemic routing). The run closes with an audit trail and a report synthesized from the committed graph.

Three things are deliberately left outside the mechanism: how many agents there are and what each specializes in, what the admission policy accepts, and what schema the committed graph uses. Those are the instantiation's, and Section 3.7 fills them in for scientific controversy review. What the mechanism fixes is the shape of the loop and the rules that govern it, and those hold for any values of the three.

Following the formalization style of executive-memory models, we make the data flow explicit. A task is a contract $X = (q, s, b, K)$ over an objective q, scope s, budget b, and optional materials K. At step t an agent issues a structured action a_t; every formal change is encoded as an event

$$e_t = (t,\ \text{actor},\ \text{type},\ \text{payload},\ \text{refs}), \quad (1)$$

and the ledger grows only by admission:

$$L_{t+1} = \begin{cases} L_t \oplus e_t, & \text{Gate}(e_t) = \text{admit}, \\ L_t, & \text{otherwise}, \end{cases} \quad (2)$$

where $\oplus$ is idempotent append keyed by a deterministic idempotency key. The evidence graph is a pure projection of the admitted ledger,

$$G = \text{Project}(L), \quad (3)$$

so replaying the ledger on an empty database reconstructs G exactly, and no other code path can write it. Each agent's working context is a role-specific sum of process recall and a projection of the committed graph,

$$C_i = \text{ProcessRecall}_i(L) + \text{EvidenceProjection}_i(G, \text{role}_i). \quad (4)$$

which is why two agents never read the same undifferentiated summary. A gap u exposed by G becomes a bounty scored on five dimensions,

$$\begin{aligned} \mathrm{b}(u) = w_1\,\mathrm{impact}(u) &+ w_2\,\mathrm{uncertainty}(u) \\ &+ w_3\,\mathrm{investigability}(u) + w_4\,\mathrm{novelty}(u) \\ &- w_5\,\widehat{\mathrm{cost}}(u). \end{aligned} \tag{5}$$

and the final round either produces a conditional consensus

$$C^* = \mathrm{JointModel}(\{C_i\}_{i=1}^n, G), \tag{6}$$

or refuses when the strongest opposing position or a falsification condition is missing. Equations (2)–(3) encode the write isolation of Section 3.4, Eq. (5) drives the routing of Section 3.6, and Eq. (6) is the no-voting rule of Section 3.3 made formal.

3.2 Generalized Executive Memory

The generalization is not a matter of instantiating MemoBrain once per agent. A single agent's memory graph answers *how this agent completed this task*; a collective's shared state has to answer *what its members now hold jointly*. Those are different questions, and reusing one graph for both breaks in five specific ways. Tool calls and committed knowledge end up in the same structure, so process and evidence cannot be told apart. One agent's mid-flight guess becomes readable as verified fact by peers who were not there when it was formed. An ordinary Fold compresses a still-open disagreement along with the settled material, which is silent deletion of dissent. Several findings inside one artifact cannot be attached to different conclusions. And no view can show how contested a conclusion currently is. Table 1 sets the two settings side by side.

Memory is organized in three layers. *Private memory* holds each agent's tool calls, failed routes, private hypotheses, and plans. *Collective executive memory* records the group's structured cognitive events: who proposed what, who attacked whom, which controversies remain open, and which gaps nobody has claimed. *The committed graph* holds what the collective has formally accepted and can be held to.

MemoBrain's native operations are redefined for the collective setting. *Quarantine* (from Flush) keeps a refuted claim instead of deleting it: the claim moves through `PROPOSED` → `SUPPORTED` → `CONTESTED` → `QUARANTINED` → `RESURRECTED/REJECTED`, and its record keeps who refuted it, what evidence is missing, and which conditions would resurrect it. *Dialectical Fold* (from Fold) compresses a dispute into a Debate Capsule only after preserving the shared understanding, the strongest supporting evidence, the strongest opposing evidence, hinge variables, scope boundaries, unresolved conflicts, and falsification conditions; the original nodes remain, and the capsule links back to them. *Perspective Recall* (from Recall) builds agent i's context as in Eq. (4): a role-specific slice of the committed graph added to that agent's process recall, so no two agents read one undifferentiated summary. The set is completed by *Fork*, which opens parallel research paths when evidence cannot distinguish competing explanations; *Resurrect*, which re-activates a quarantined hypothesis once new evidence meets its resurrection conditions; and *Merge*, which records conditional merge candidates for the operator to adjudicate.

Two of these operations carry a requirement with no counterpart in the single-agent setting, and between them they account for most of the architecture that follows. *Memory is public.* A lone agent that compresses badly pays for it itself; in a collective, a compression one member performs is read by members who did not perform it, so compression stops being a private optimization and becomes an edit that has to stay reconstructible. *Process must separate from commitment.* The ambiguity that costs a lone agent a wasted step costs a collective a fact it believes, because peers cannot tell a working guess from an audited conclusion unless the two live in structures with different write rules.

3.3 The Collective Protocol

The memory layer only works if the agents produce material it can govern. They submit structured actions; free text is rejected: `PROPOSE`, `SUPPORT`, `CHALLENGE`, `QUALIFY`, `FORK`, `REQUEST`, `REVISE`, `DISSENT`; when challenged they may `DEFEND`, `REVISE`, `NARROW`, `WITHDRAW`, or `DISSENT`. Every action carries an actor, a target, a statement, evidence references, a confidence, a falsification condition, and a novelty mark. The ledger rejects factual assertions without evidence, target-less comments, duplicated positions, and agreement that adds no information. None of the action names is domain-specific: each

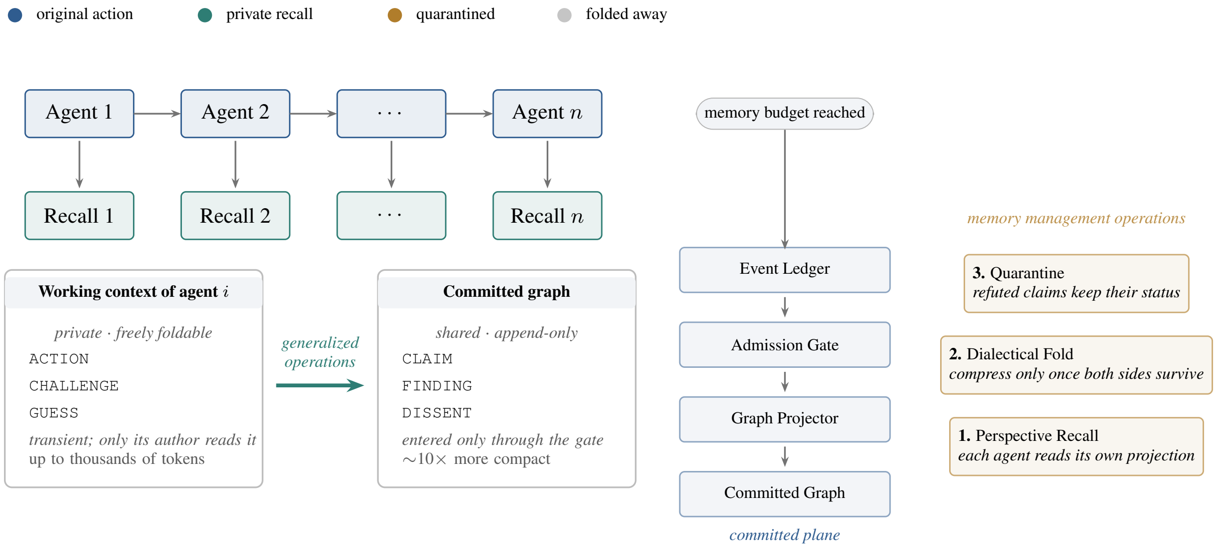

Figure 1: EpistemoBrain. Top: an agent collective of n role-specialized members acts in rounds; each member keeps a private recall (teal) over its own process memory. Bottom left: the zoom-in contrasts what one member holds against what the collective holds. A working context is transient, private and freely foldable; the committed graph is shared, append-only, and roughly an order of magnitude more compact, and material enters it only through the pipeline on the right. Bottom right: the three generalized operations that turn the one into the other, and the committed plane every formal change must pass through. The generalization is not a per-agent instantiation of MemoBrain: the operations act on a shared state that agents who did not write it will read.

describes what one member does to another member's commitment.

Agents gather evidence without seeing one another and commit positions first; evidence is exchanged only afterward, so cross-examination examines genuinely independent positions instead of restating a shared draft. The reference instantiation runs seven rounds, though the round list is a parameter of the instantiation, not of the mechanism: independent precommitment, professional acquisition, evidence exchange, cross-examination, gap bounty, joint modeling, final rejudgment.

No majority vote decides what the collective commits. The joint-modeling round refuses to produce a consensus when the strongest opposing position or a falsification condition is missing. An agent that cannot accept the consensus receives a Dissent Certificate, a first-class node in the committed graph that records what it opposes, why, and what evidence would retract the dissent.

An operator may also intervene directionally at one fixed point, after the gap bounty and before joint modeling: the run pauses, presents the agents' precommitment positions and confidence levels, and accepts a short directional note, or an explicit decision to leave it blank. Both paths are equally valid. The note enters joint modeling only under the label "operator directional note, not a judgment by a participant". It never changes an admission decision, never serves as evidence for or against a claim, and never enters a Debate Capsule or Dissent Certificate. Identical input must produce identical admissions, dissents, and gate decisions with or without a note, a property asserted by tests. The checkpoint is directional control, not an extra vote.

3.4 Dual-Graph Governance

Process and commitment are stored apart. The *Process Graph* answers "how did the agents work": tasks, tool calls, failed routes, challenges, and decisions, managed by MemoBrain and freely foldable. The *Committed Graph* answers "what do we now hold": the audited objects the collective is prepared to defend, in a schema the instantiation supplies (Section 3.7 gives the scientific one).

The two graphs are coupled by the *Event Ledger*. Every formal change is first appended as an idempotent, replayable, auditable event, and the *Graph Projector*, the ledger's only consumer and the committed graph's only writer, applies the admission gate and materializes the admitted nodes and edges (Eqs. (2)–(3)). Write isolation is enforced at the database privilege level, not by coding discipline. The application role holds no insert, update, or truncate privilege on the committed-graph tables, and the projector role holds no privilege on the ledger,

	MemoBrain (one agent)	EpistemoBrain (a collective)
Unit of analysis	the agent's own reasoning trajectory	the collective's shared cognitive state
Memory owner	private to the agent that wrote it	public; read by agents that did not write it
Flush	prune an invalid or failed path	*Quarantine*: keep the refuted claim, its refuter, and the conditions that would revive it
Fold	compress a completed sub-trajectory	*Dialectical Fold*: compress a dispute only after both strongest positions, the hinge variables, and the falsification conditions survive
Recall	rebuild a high-salience context for the agent	*Perspective Recall*: a role-specific projection of the committed graph per agent
Graph separation	process and commitment may share one graph	forced into two graphs: process is freely foldable, commitment is append-only
Aggregation	not applicable to one agent	no majority vote; conditional consensus with named dissent

Table 1: What changes when executive memory is generalized from one agent to a collective. Each row is a constraint the single-agent setting does not raise, and each is the reason the corresponding mechanism takes the form it does.

so a bug on either side cannot forge the other's data. Refuted, quarantined, and folded nodes are never physically deleted; they change status and remain traceable.

The separation is what makes the memory safe to compress. Because the process graph is never read as commitment, an aggressive Fold there costs nothing but recomputation. Because the committed graph is append-only, no compression anywhere can remove a record that a later agent, or a human reviewer, needs in order to reconstruct why something is believed.

3.5 The Admission Gate

Nothing reaches the committed graph except through the gate, and the gate is where an instantiation exercises most of its control. Structurally it is a staged pipeline: a candidate must clear schema validation, semantic deduplication, verification against the source material, entailment (does the source actually support the claim drawn from it), a quality stage, and a graph-consistency stage. The stages are fixed by the mechanism; what each one checks is supplied by the instantiation, and a candidate that fails any stage is quarantined with an audit record instead of being silently dropped.

Two properties of the gate matter more than its particular stages. First, the gate is applied by the projector, not by the agents: one component proposes and a separate one decides, so no participant can admit its own claim. Second, grading is explicit and the bar is monotone. Material is graded by how directly it can be checked, and the strongest conclusions are barred from resting on a weaker grade no matter how many weak sources agree. That is the structural answer to the shared-error failure mode, because repetition of something that cannot be independently checked never accumulates into strength.

The gate also asks whether apparently multiple sources are actually one. Lineage detection groups sources that share a dataset, an overlapping sample, a preprint lineage, an extension, a reanalysis, or a meta-analytic inclusion, and the system reports the raw count alongside the independent-cluster count: six sources resting on one dataset count as one independent cluster.

Four mechanisms defend against the subtler failure of *shared* errors among the agents themselves. *Blind review* strips prestige signals (author, venue, citation counts) from first substantive judgments. *Independent dual extraction* extracts high-impact material twice through separate paths and escalates discrepancies. *Diversity constraints* raise a gap automatically when all support for a claim traces to one dataset, team, jurisdiction, or design. *Adversarial retrieval* issues reverse queries against the sources of each confirmed claim, looking for refutation, null results, alternative explanations, measurement criticism, replication failure, and boundary inversion.

3.6 Gap-Driven Routing

The collective is not a scripted play-through of a fixed round list. Gaps the committed graph exposes generate bounties (Eq. (5)), scored on impact, uncertainty, investigability, novelty, and cost, then ranked and broadcast for agents to claim. What sets the next round of investigation is the state of what the collective has committed, not a moderator reading from a script. Compressing memory and directing work are therefore the same operation

Role	Responsibility
Theory Builder	mechanisms, competing explanations
Causal Scientist	confounding, reverse causation, selection
Measurement Scientist	operationalization, self-report bias
Replication Scientist	power, sensitivity, replication history
Boundary Scientist	population, culture, platform limits
Adversarial Falsifier	strongest counterexamples
Evidence Auditor	provenance, entailment, independence

Table 2: The seven roles of the reference instantiation and their primary responsibilities.

seen from two sides: a Fold that dropped an unresolved conflict would silently delete the work item that conflict represents, which is one more reason Dialectical Fold has to preserve it.

A gap is a first-class object rather than a limitations paragraph, and that changes what a stalled investigation can hand back. When a bounty round closes, the system may propose a *DiscriminatingStudy*: a design whose competing predictions would separate two still-live explanations, with its expected information gain attached. The output of an investigation that could not conclude is a specification for the next one, which is something a summary cannot produce.

3.7 Reference Instantiation: Scientific Controversy Review

The instantiation we build and measure is a scientific controversy review: the task is a contested question, the materials are the literature, and the committed graph is an evidence graph. It is a demanding testbed, not the definition of the mechanism, and the three things the mechanism leaves open are filled in as follows.

Roles. Seven specialist agents, one for each question a contested claim has to survive. Table 2 lists them. Each has an independent role specification, independent private state and private memory, and a different projection weighting over the committed graph.

Rounds. Seven. (1) *Independent Precommitment*: agents commit initial judgments and confidence levels without seeing one another (sealed first, read later). (2) *Professional Acquisition*: agents issue evidence requests from their own perspectives; a query planner merges them and executes against shared sources (OpenAlex, Crossref, Semantic Scholar, Unpaywall, and operator-supplied materials). (3) *Evidence Exchange*: structured evidence is exchanged, never private reasoning histories. (4) *Cross-Examination*: targeted challenges on causality, measurement, statistics, boundaries, sources, and hidden assumptions. (5) *Blindspot Bounty*: gaps the evidence graph exposes are scored on impact, uncertainty, investigability, novelty, and cost, then ranked and claimed. (6) *Joint Modeling*: a conditional consensus is drafted together with the shared understanding, the strongest supporting and opposing positions, hinge variables, scope boundaries, unresolved conflicts, and falsification conditions. (7) *Final Rejudgment*: agents judge independently again, recording belief updates and named dissents.

Gate contents and schema. Evidence is graded A through D. Level A requires full text with exact quotes and precise location; Level B is abstract plus metadata; Level C covers second-hand descriptions in reviews; Level D covers web or news leads. Levels B, C, and D can never alone support a high-confidence causal conclusion. Three audits apply to formal findings: source authenticity (DOI, title, authors, version, retraction, PDF identity), citation entailment (whether a quote supports the claim, whether correlation is reported as causation, whether qualifiers were dropped), and method quality (directness, design strength, measurement validity, statistical precision, replicability, external validity). A *CausalUpgradePolicy* intercepts the combination of a cross-sectional design with a causal claim at the consistency stage and quarantines the event with an audit record. The committed graph holds ten node types (ResearchQuestion, Claim, Source, StudyFinding, Construct, Context, Blindspot, DebateCapsule, DiscriminatingStudy, DissentCertificate) and twelve edge types (SUPPORTS, REFUTES, QUALIFIES, CONTRADICTS, CONFOUNDS, MEDIATES, MODERATES, OPERATIONALIZES, DERIVED_FROM, APPLIES_IN, EXPOSES, TESTS).

Vocabulary. Gaps are called blindspots here, and bounties over them are what the evaluation measures. The committed graph is an evidence graph, the ledger is a scientific event ledger, and the operator is a researcher. None of these names changes the mechanism.

3.8 Engineering Realization

We implement the reference instantiation as **Poliscope**, an open-source system whose repository also carries the evaluation harness. The mechanism is realized as a modular monolith with an asynchronous worker and PostgreSQL as the source of truth; no workflow state lives only in memory. The worker claims tasks with `SELECT ... FOR UPDATE SKIP LOCKED`. A run executes each phase in one transaction as the application role and commits; the projector then runs in a second transaction as the projector role. Two identities, two transactions, one ledger. All model calls go through a model gateway (OpenAI-compatible endpoints, streaming with a non-streaming fallback, per-task model configuration, cost and latency accounting), and all external tools go through a tool gateway (OpenAlex, Crossref, Semantic Scholar, Unpaywall, operator knowledge bases). A single agent's failure degrades the round: the agent is marked absent with a reason on the ledger instead of aborting the task. Structured-output repair failures are quarantined and never write the committed graph. Live views are served as two SSE streams, the formal ledger stream and a best-effort process stream for token-level progress, so the audit trail is never polluted by reasoning noise. The implementation is at `https://github.com/Fishman-free/poliscope`, archived at `https://doi.org/10.5281/zenodo.22541350`.

4 Experiments

Everything in this section measures the reference instantiation of Section 3.7, not the mechanism in the abstract. The three parameters the mechanism leaves open (roles, admission policy, graph schema) take their scientific values here, and the numbers below are properties of that setting. The mechanism-level claims of Section 3 are structural; the experiments test whether the structure holds up when it is actually run.

4.1 The ForesightBlindspot Benchmark and Baselines

We design **ForesightBlindspot**, a time-sliced evaluation protocol for blindspot discovery in computational social science (domain: digital behavior, social media, and mental health). Each case is a contested question with a cutoff date: the system may read only the pre-cutoff corpus. Gold blindspots are gaps that later high-quality research confirmed, that had reasonable premonitory evidence before the cutoff, and that do not depend on entirely unforeseeable platforms, events, or methods. Leakage controls are built into the protocol: closed pre-cutoff corpora, no future citations, anonymizable variables, and an explicit disclosure that base-model parameters may contain future knowledge, a risk that cannot be fully eliminated. The metric family covers blindspots (recall, precision, high-impact recall, specificity, actionability, foresight distance), evidence (citation existence, citation entailment, evidence independence, causal-overclaim rate, full-text coverage), memory (compression ratio, scientific-backbone retention, contradiction retention, long-horizon drift, recovery quality), collaboration (role diversity, debate utility, belief-update quality, dissent preservation, false-consensus rate), and cost (cost per valid blindspot, tokens per admitted finding).

The five baselines are configurations of the real collective, not separate implementations, so each adds exactly one capability over the previous: (1) *Single-Agent Deep Research*, one agent (the theory builder) with no protocol, no memory, and no gate; (2) *Fixed Multi-Agent Debate*, seven agents with an undifferentiated deliberator, no per-agent memory, and no gate; (3) *Collective + Linear Context*, seven agents sharing one undifferentiated transcript; (4) *Collective + MemoBrain without Evidence Engine*, seven agents with per-agent memory but the gate removed; and (5) *Full Poliscope*, the complete system.

4.2 Baseline Construction Correctness

Before reporting numbers, we validated that the baselines are what they claim to be, and the first evaluation run exposed two construction defects. First, the single-agent baseline selected its role as the alphabetically first role of an ordered enumeration, which happened to be the adversarial falsifier. The "lone researcher" was in fact the falsifier, and in our scripted material the falsifier holds the strongest blindspot answers, so the single agent scored as high as the full collective (F1 0.8 on all five variants). Second, the final-rejudgment handler iterated all seven agents unconditionally, minting placeholder judgments for agents that never participated. We corrected both: the single-agent baseline explicitly uses the theory builder, and the rejudgment handler respects the participating-agent set with a compatible fallback that leaves the seven-

Baseline	Rec.	Prec.	F1
Single-Agent Deep Research	0.143	0.500	0.222
Fixed Multi-Agent Debate	**1.000**	**0.778**	**0.875**
Collective +Linear Context	**1.000**	**0.778**	**0.875**
Collective +MemoBrain, no gate	**1.000**	**0.778**	**0.875**
Full Poliscope	**1.000**	**0.778**	**0.875**

Table 3: Blindspot recall, precision, and F1 on the scripted demo case (B_dev). Best per column in bold. All four seven-agent variants find all seven gold blindspots and are tied on every blindspot metric on this material; the single agent, restricted to the theory builder, finds one.

agent production path byte-for-byte unchanged. A regression guard now asserts that single-agent blindspot coverage is strictly below multi-agent coverage, so a future regression that collapses the baseline ladder fails immediately. We report this correction because the construction of the measurement instrument is part of the measurement.

4.3 Main Results on the Scripted Case

Table 3 reports the five baselines on a scripted demo case (screen time and adolescent depressive symptoms) in which every gateway is deterministic: no model calls, no network, no database. Each agent nominates one specialist blindspot in the bounty round (the falsifier nominates two, one deliberately duplicating the boundary specialist's nomination to model overlapping multi-agent nominations). Gold blindspots are one keyword per specialist role.

Blindspot coverage is driven by *agent coverage*. The lone agent finds one of seven specialist blindspots (F1 0.222) while every seven-agent variant finds all seven (F1 0.875), so the collective's contribution is large and it belongs to the set of roles, not to any one role in particular. After the construction fix, the single agent's score follows from what its own role contributes, not from whether the falsifier happens to be present. The four seven-agent variants then score identically, which is a structural consequence of the harness: every variant collects all seven agents' outputs, the scripted gateway ignores prompts and memory, and the gate admits bounty events unconditionally. What the protocol, the memory layer, and the evidence gate add to blindspot *integration* therefore leaves no trace on this metric when the material is scripted. Those contributions show up in the auxiliary scores (dissent preservation, citation entailment, evidence independence), and observing them on blindspots themselves requires real models. We treat the null result as an identified boundary of the measurement.

4.4 Ablation Study

Beyond the five-baseline ladder, we ablate the full system one capability at a time (design spec 11.4): six *full minus X* variants, each removing exactly one mechanism while everything else, including the evidence gate, stays intact. The removals are (1) independent precommitment (the phase does not run); (2) the adversarial falsifier role; (3) the evidence auditor role; (4) the dialectical fold (JOINT MODELING skips the DebateCapsule and records an unfilled slot instead of silently dropping the opposition); (5) evidence lineage (acquired sources arrive without dataset metadata, so evidence independence cannot see that two papers share one dataset); (6) MemoBrain (per-agent private memory is replaced by one undifferentiated transcript). Each ablation is a configuration of the same orchestrator the full system runs. The switch is one flag or one agent subset, not a reimplementation, so the contrast with Full Poliscope isolates that capability's contribution.

Table 4 shows three mechanism-level effects. Removing the adversarial falsifier drops recall from 1.000 to 0.857: the publication-bias blindspot was its nomination, and no other role covers it, so a collective without a dedicated falsifier misses exactly the gap that role exists to find. Precision rises at the same time (0.778 to 0.857), because the falsifier's second nomination deliberately duplicates the boundary scientist's, and without the falsifier no statement is double-paid. Removing the evidence auditor also costs recall (1.000 to 0.857, F1 0.800): the provenance blindspot belonged to that role. Removing lineage raises evidence independence from 0.667 to 1.000. Without dataset metadata, two papers that share a cohort are counted as independent evidence, so the number the system reports about its own evidence quality is *inflated by the absence of the very mechanism that would correct it*. The dialectical-fold ablation cannot move the blindspot metric, but it is visible in the honesty signal: the DebateCapsule disappears and the run records one additional unfilled slot instead of pretending the debate had no opposition. Dissent preservation is a property of the protocol, and its absence is recorded, not hidden.

Two ablations show no effect on this scripted material, and we say so plainly instead of manufac-

Variant	Rec.	Prec.	F1	Indep.	Unfilled
Full Poliscope	**1.000**	0.778	**0.875**	0.667	1
− Precommitment	**1.000**	0.778	**0.875**	0.667	1
− Adversarial falsifier	0.857	**0.857**	0.857	0.667	1
− Evidence auditor	0.857	0.750	0.800	0.667	1
− Dialectical fold	**1.000**	0.778	**0.875**	0.667	2
− Lineage	**1.000**	0.778	**0.875**	1.000†	1
− MemoBrain	**1.000**	0.778	**0.875**	0.667	1

Table 4: Ablations of the full system on the same scripted case (B_dev). Best per column in bold. "Indep." is evidence independence; "Unfilled" counts slots the run records as unfinished instead of fabricating them. Each ablation removes exactly one capability; the adversarial falsifier and the evidence auditor each carry a gold blindspot of their own, which is why removing them costs recall. † Higher is worse here: removing lineage tracking strips dataset metadata, so sources that share a cohort are miscounted as independent and the system's self-report is inflated to a perfect 1.000 by the absence of the mechanism that would correct it.

turing a difference. Removing precommitment removes the seven PRECOMMITMENT_SEALED events but leaves every score unchanged; removing MemoBrain changes nothing either. This is the same structural boundary identified for the baseline ladder: a scripted gateway answers identically regardless of prompt, memory, or phase history, so protocol-level integration effects (what precommitment changes about later reasoning, what per-agent memory changes about what an agent recalls) are not observable without real models. The three ablations that do show effects (falsifier, auditor, lineage) do so because they change a *participant* with scripted content or the data a scoring function sees, not because the mechanism's behavioral consequences are being measured. We therefore read the three effects as validated mechanism checks and the two null results as an identified measurement boundary, not as evidence that the two capabilities contribute nothing.

4.5 Real-Model Evaluation and the Semantic Judge

The scripted case is deterministic, and its blindspot metric rests on a *keyword* proxy: "reverse_causation" matches only statements that literally contain those words (underscore-separated and case-folded). The scripted gateway writes its statements to contain the keyword, so the proxy works there. It does not survive contact with a real model. Free-form output such as "the direction of effect could run the other way" names the reverse-causation gap without containing the token, so the keyword score reports recall 0.0 even when the gap was found. To measure blindspot quality under a real model we added a *semantic judge*: for each gold concept, the same model gateway (no second vendor SDK) is asked which numbered candidate statements semantically address that concept, and recall and precision are derived with the identical definitions the keyword score uses. A judge that is unreachable, or whose structured answer is quarantined, returns *unmeasured* instead of a silent 0.0.

We then ran five key variants (the full system, the single-agent baseline, and the three mechanism ablations whose scripted effects were largest) through the real gateway (DeepSeek-V4-Flash on the official DeepSeek API), three runs each, and scored blindspots with the semantic judge. An initial eleven-variant single-run sweep was abandoned after the first endpoint rate-limited for two days and polluted twelve runs (unfilled slots jumped to 46, and the judge was unreachable for every semantic call); those runs were discarded, never reported. Table 5 reports the three-run means (F1 is recomputed from mean recall and precision, not averaged over per-run F1s).

The semantic judge repairs the measurement. Where the keyword proxy reports recall 0.0 for the full system on free-form text, the judge produces a readable, non-degenerate distribution. Repeated runs then turn the earlier single-run reading ("sampling variance dominates") into a stable *precision–recall trade-off*. Full Poliscope achieves perfect precision (1.000) at the cost of the lowest recall (0.286: exactly two of seven gold concepts, in all three runs). Every simplified variant, the lone researcher included, finds nearly all gold concepts (recall 0.809–1.000) at markedly lower precision (0.406–0.586). Removing precommitment reproduces recall 1.000 in every run, so the earlier single-run observation was a real, repeated effect and not a sampling artifact. Unfilled slots explain the gap. The full system leaves 12.3 slots

Variant	Sem-R	Sem-P	Sem-F1	Unf.	n
Single-Agent Deep Research	0.809	0.544	0.651	1.0	3
Full Poliscope	0.286	**1.000**	0.444	12.3	3
– Precommitment	1.000	0.490	0.658	1.0	3
– Adversarial falsifier	**1.000**	0.406	0.577	1.7	3
– Evidence auditor	0.952	0.586	**0.726**	1.3	3

Table 5: Blindspot scoring on repeated real-model runs, means over $n = 3$. Best per column in bold. "Sem" is the semantic judge; R/P/F1 are recall, precision, and F1; "Unf." is the mean number of unfinished evidence slots. The keyword proxy is omitted: on this endpoint it again collapses to 0 for the full system, and the semantic judge is the controlled measurement. Full Poliscope is the only variant with perfect precision, and it pays for that precision in recall and in unfilled protocol slots.

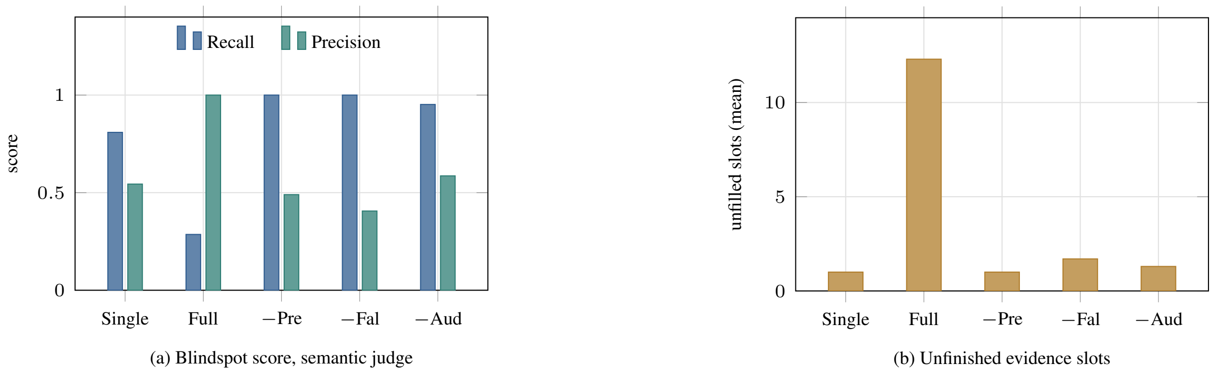

Figure 2: Repeated real-model runs ($n = 3$ per variant), means. Panel (a) separates the two failure modes the single F1 column conflates: the full system attains precision 1.000 while the simplified variants trade it away for coverage. Panel (b) shows the mechanism behind the full system's low recall: the same governance that lifts precision also raises the number of protocol slots the model leaves unfilled, from 1.0 to 12.3 on a flash-tier endpoint. "Single" is Single-Agent Deep Research; "Full" is complete Poliscope; the remaining three remove one component from the full system (precommitment, adversarial falsifier, evidence auditor).

unfilled on average when the model produces no structured nomination, against 1.0–1.7 elsewhere, so protocol burden on a flash-tier model is the binding constraint: governance buys precision at the cost of recall, and under a weak model the recall cost dominates F1. The deterministic data-level scores (citation entailment 0.5, evidence independence 0.667, dissent preservation 1.0) are identical across all fifteen runs, as they must be for scoring that never reads model output.

4.6 Analysis: What the Trade-off Says About Deployment

The real-model results are not a negative result, and reading them as one would miss the paper's central design claim. The governance mechanisms EpistemoBrain adds operate at the *integration* level: they shape what agents reason and recall across rounds, and their benefit needs a model strong enough to sustain the full protocol without the agent absence observed here (12.3 unfilled slots against 1.0–1.7). What the repeated runs establish is a deployment map. When evidence quality is the priority (a contested review where an admitted causal claim can be wrong in a costly way), the full protocol is the only configuration with perfect blindspot precision and intact honesty signals: dissent preservation 1.0 across all fifteen runs, citation entailment and evidence independence computed from data, not asserted. When coverage is the priority and the available model is a flash-tier one, a lighter protocol (or precommitment removed) recovers recall 0.809–1.000 at a known, quantified precision cost. The scripted case isolates each mechanism's structural contribution; the real-model runs locate the model-strength boundary at which that isolation stops. Both pieces are needed to choose a configuration honestly, and the system reports whichever side of the trade-off it is running on instead of averaging the two into a single score.

4.7 Cost and Protocol Burden

Agent coverage costs something, and the ledger makes the cost countable. Every formal change

Variant	Events	Rec.	Ev./blindspot
Single-Agent Deep Research	34	0.143	34.0
Full Poliscope	60	**1.000**	**8.6**
− Precommitment	50	1.000	7.1
− Adversarial falsifier	54	0.857	9.0
− Evidence auditor	56	0.857	9.3
− Dialectical fold	59	1.000	8.4
− Lineage	60	1.000	8.6
− MemoBrain	60	1.000	8.6

Table 6: Ledger volume against blindspot yield on the scripted case (B_dev). "Ev./blindspot" divides ledger events by the number of gold blindspots found (recall $\times$ 7); lower is cheaper. The full system writes $1.8\times$ the events of the lone agent and returns $7\times$ the blindspot coverage, so its cost *per* blindspot is about four times lower.

is an event, so the ledger length of a run is a direct measure of how much protocol the run actually executed. Table 6 sets that volume against the blindspot yield.

The lone agent is the cheapest run in absolute terms (34 events) and by far the most expensive per result (34.0 events per gold blindspot, against 8.6 for the full collective). Adding agents does not multiply cost the way it multiplies coverage: a seven-agent run writes $1.8\times$ the events of a one-agent run while returning $7\times$ the blindspots. The reason is structural rather than economical. Agents share one runtime, one retrieval cache, and one ledger, so the marginal agent adds its own structured actions and nothing else. Budget control in the implementation targets the same quantity this table measures: speaking budgets, semantic deduplication, and shared retrieval bound ledger growth without dropping an agent.

The real-model runs invert part of this picture, and the inversion is the same measurement boundary the previous sections identify. Under a flash-tier endpoint the full system wrote *fewer* events (118.3 on average) than every ablated variant (148.0–160.3), because an agent that cannot produce a schema-valid contribution also fails to emit the challenges, nominations, and judgments that would have followed from it. Protocol volume and recall move together here. The full system is not paying more for less; it is executing less of its own protocol, and the unfilled-slot count (12.3) is where the missing work is recorded, not hidden.

4.8 System Verification

Beyond the controlled comparison, Poliscope is verified end to end. The test suite comprises 752 unit tests and 364 integration tests over a real PostgreSQL instance with three database identities, including privilege tests that assert the application role cannot write the evidence graph and the projector role cannot rewrite the ledger: dual-graph isolation is a database fact, not a code convention. Real research tasks run through the full seven-agent collective on real OpenAI-compatible endpoints, producing conditional consensus, Debate Capsules, Dissent Certificates, and a synthesized final paper whose references are bound to DOIs. Tasks that exhaust budget or lose agents finish as "completed with gaps", with every absent agent and unfilled slot named in the report, never fabricated into a success. The event ledger is idempotent and replay-safe (re-running a task is a no-op), and the streaming live view survives reconnection via Last-Event-ID.

5 Conclusion

We presented EpistemoBrain, which generalizes executive memory from a single reasoning agent to an agent collective. The unit of analysis moves from a private trajectory to a shared cognitive state, and each native operation changes meaning with it: Quarantine keeps a refuted claim and its history, Dialectical Fold compresses a dispute only after both positions and the falsification conditions survive, and Perspective Recall gives each agent its own projection of the shared state. That shift forces the rest of the architecture, and it does so without naming a task domain. Because a collective's memory is read by agents that did not write it, process and commitment are separated into two graphs joined by an append-only ledger with database-enforced write isolation, an admission gate checks every formal change before it becomes a conclusion, and a process note stays permanently distinguishable from commitment. Dissent is never deleted; where the collective cannot agree, the outcome is conditional consensus plus a named certificate. The design thesis is that a collective's reliability is a memory-and-governance problem rather than a prompt problem, and that is why the mechanism transfers across settings where prompt engineering does not.

The ForesightBlindspot protocol, the five-baseline comparison, and the six-variant ablation ladder establish a measurement basis for one instantiation, with one open caveat: scripted gateways make the protocol's integration-level contributions unobservable on the blindspot metric, which is exactly why two of the six ablations are null on this material. The real-model study shows that a keyword proxy for blindspot recall fails on free-form text and that a semantic judge repairs the measurement; repeated runs then expose a stable deployment boundary, not noise. Under a flash-tier model, the full collective buys perfect blindspot precision at the cost of recall, and agent absence (12.3 unfilled slots against 1.0–1.7) is the binding constraint – the memory mechanism cannot demonstrate its coverage benefit when the model cannot fill the slots it manages. Observing those benefits therefore requires either a stronger model or a deliberately lighter protocol, and the system is built to report which regime it is running in. The generality claim itself rests on construction, not on this measurement: the mechanism's definitions mention agents, a shared graph, and an admission policy, and nothing else. Running a second instantiation is the experiment that would convert that structural argument into an empirical one.

Limitations

The boundaries of this study are as much a part of its result as the measurements are, and each one below names the experiment that would close it.

Generality is argued, not measured. The mechanism is stated over agents, a shared committed graph, and an admission policy, and none of those three names a domain – but that is a property of the definitions, not a result. Everything in Section 4 comes from one instantiation, a scientific controversy review, so those numbers establish that the structure holds when it is run, not that it transfers. Supplying a second instantiation means supplying three things (roles, admission policy, graph schema), which is the direct test, and nothing in the present study rules out that some of what we call mechanism is in fact an artifact of the scientific setting.

Evidence ceiling. The acquisition pipeline reliably obtains bibliographic metadata and abstracts, so the strongest evidence the system admits unaided is Level B. Full-text retrieval with exact, page-anchored quotes is implemented but not yet complete across the source corpus, and Level B evidence is by design barred from supporting a high-confidence causal conclusion on its own. The system therefore holds itself to a stricter standard than its own evidence supply currently meets, which is conservative but caps the strength of any single admitted causal claim.

Dependence on model strength. The integration-level benefits of the protocol surface only when the underlying model can carry a schema-valid contribution through all seven rounds. On the flash-tier endpoint used here, protocol burden, not governance quality, is the binding constraint: the full system leaves 12.3 evidence slots unfilled against 1.0–1.7 for the simplified variants, and that absence is what depresses its recall. The precision it buys is real and reproducible, but the coverage side of the trade-off is measured under a model that is not strong enough to exercise the mechanism, so the coverage claim remains open. Repeating the ladder on a reasoning-tier endpoint is the direct test.

Instrument still under construction. ForesightBlindspot's time-sliced closed corpus and double-annotated human gold standard are not yet produced. The blindspot metric reported here rests on a scripted demo case whose statements are written to contain the gold keyword, plus a semantic judge over free-form output. The keyword proxy collapses to zero recall on real text, which is precisely why the semantic judge was added; until human labels exist, every number should be read together with the scoring protocol that produced it.

Single case, no held-out split. The comparison runs one scripted case and one repeated real-model case. There is no independent test split, so the ablation ladder functions as a mechanism check on controlled material not a generalisation claim about contested questions at large.

Upstream memory integration. The memory layer runs against an in-memory adapter and a simplified fold behind a frozen three-layer interface. Substituting the upstream implementation requires no change to the collective or gate layers, but the long-horizon anti-drift benefit of executive memory currently rests on a structural guarantee, not on a cross-task measurement.

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A Appendix: Implementation Notes

Deterministic evaluation harness. The Foresight-Blindspot harness (`packages/evaluation/`) runs the five baselines and the six ablation variants as configurations of the real orchestrator against a database-less event sink, so a run is deterministic and fast enough for unit tests; scoring functions reuse the production audit rules (causal upgrade policy, citation entailment verifier, lineage detection and evidence clustering) instead of re-deriving them, so a variant's score reflects the same standard the gated system is held to. Each ablation is one protocol switch on that same orchestrator – a phase subset, an agent subset, a fold flag, or a lineage-free acquirer – never a reimplementation (Section 4.4). Gold blindspot matching is implemented as a keyword proxy, with a double-annotation agreement protocol (Kappa/Alpha statistics, adjudication) in place.

Permissions as design. PostgreSQL grants implement the write isolation: the application role can append to the ledger and read the evidence graph but cannot insert, update, or truncate graph tables; the projector role can write the graph and advance event status but cannot rewrite event payloads; neither role can truncate anything. A documented exception grants the application role DELETE for the session-lifecycle operation of destroying a whole task at the researcher's explicit request – the one path by which evidence rows may be physically removed.

Reproducibility. The repository pins migrations (Alembic), seeds, and Docker Compose services (postgres, migrate, api, worker, web, caddy); the demonstration task seed runs the real orchestrator with scripted model and tool stand-ins, and the evaluation harness runs with scripted gateways, so all numbers in this paper are reproducible without external credentials.